E-COMMERCE ANALYTICS PLATFORM (BRAZIL)

➢ Business Problem Statement :-

The company is a large e-commerce marketplace where:

- Customers place orders
- Sellers fulfill products
- Payments happen via multiple methods
- Customers provide reviews
- Logistics teams manage deliveries
- Data arrives daily as CSV files from different source systems.

➢ Business Challenges :-

- Data comes from multiple systems in raw CSV format
- No historical tracking of seller changes
- No incremental loading (full reload every time)
- No unified analytics model
- Reporting is slow and unreliable

➢ (Business Requirements)

- **Business wants a daily automated data platform that can:**

Business Questions-

- **How much revenue was generated daily?**
- **Who are the top sellers?**
- **Which product categories perform best?**

For any queries please contact. Rahul M (8551853004)

- What is the average delivery time?
- How satisfied are customers (reviews)?
- Track seller location change s over time
- Support historical reporting

End Goal (Final Output)

- Fully automated daily pipeline (7 AM)
- Clean & trusted data
- Incremental processing
- SCD-1 and SCD-2 implemented
- Star schema (Fact & Dimensions)
- BI-ready Gold tables

Technology Stack

Cloud – Azure Databricks

Engine - Apache Spark

Language - PySpark

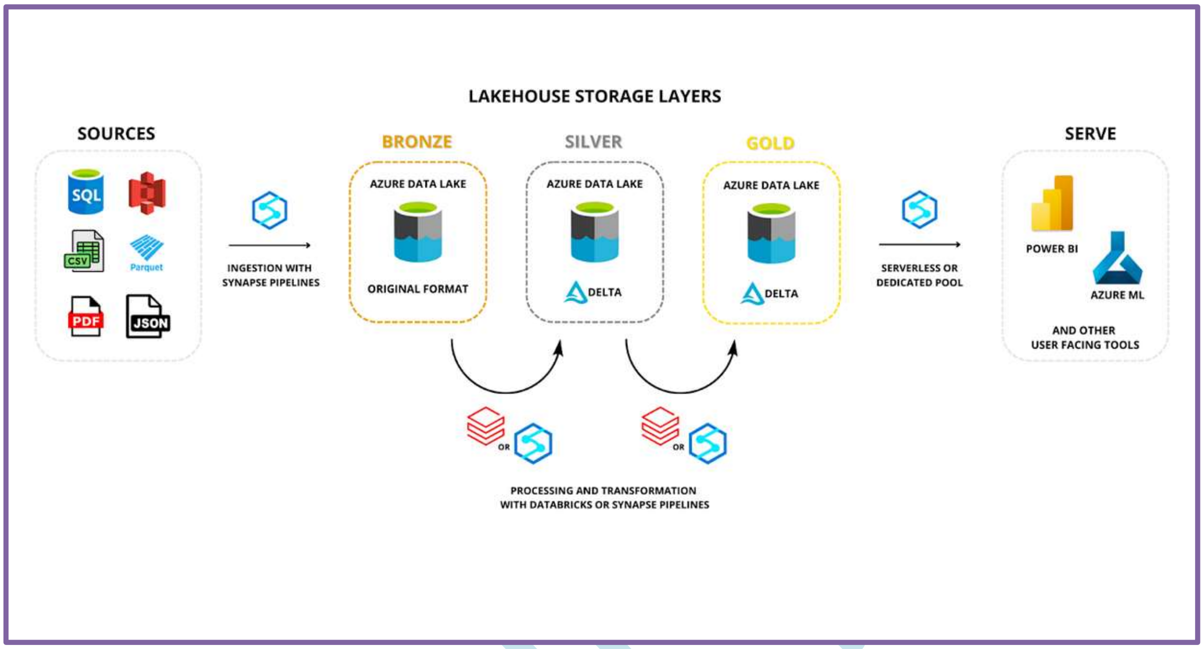
Storage - Delta Lake

Orchestration - Databricks Workflows

Format - CSV → Delta

Scheduling -Daily at 7 PM

Architecture -Medallion



PROJECT ARCHITECTURE

DATASETS :-

Customers

Column Name	Data Type	Description	Action
customer_id	string	Unique customer identifier	PK
customer_unique_id	string	Unique person (can have multiple orders)	Keep
customer_zip_code_prefix	int	ZIP prefix	Remove spaces between zip codes
customer_city	string	Customer city	Lowercase in Silver
customer_state	string	State code	Capitalize
ingest_ts	timestamp	Ingestion timestamp	Add timestamp column

Orders

Column Name	Data Type	Description	Action
order_id	string	Unique order id	PK
customer_id	string	Customer reference	FK
order_status	string	delivered, shipped, etc	Filter
order_purchase_timestamp	timestamp	Order created time	Cast
order_approved_at	timestamp	Approved time	Nullable
order_delivered_carrier_date	timestamp	Shipped date	Nullable
order_delivered_customer_date	timestamp	Delivered date	Nullable
order_estimated_delivery_date	timestamp	Estimated delivery	Nullable
ingest_ts	timestamp	Ingest time	Add timestamp column

Order_items

Column Name	Data Type	Description	Action
order_id	string	Order reference	FK
order_item_id	int	Item sequence	Keep
product_id	string	Product reference	FK
seller_id	string	Seller reference	FK
shipping_limit_date	timestamp	Shipping SLA	Cast
price	double	Item price	Remove $ symbol
freight_value	double	Shipping cost	Remove $ symbol
ingest_ts	timestamp	Ingest time	Add timestamp column

Payments

Column Name	Data Type	Description	Action
order_id	string	Order reference	FK
payment_sequential	int	Sequence number	Ignore
payment_type	string	credit_card, boleto	Remove special chars like @,# use only _ like credit_card not credit@card or credit#card
payment_installments	int	Installments count	Aggregate
payment_value	double	Amount paid	Remove Null and Any Special Chars
ingest_ts	timestamp	Ingest time	Add timestamp column

Reviews

Column Name	Data Type	Description	Action
review_id	string	Review id	PK
order_id	string	Order reference	FK
review_score	int	Rating (1–5)	Default 0
review_comment_title	string	Review title	Optional
review_comment_message	string	Review text	Optional
review_creation_date	timestamp	Review date	Cast
review_answer_timestamp	timestamp	Seller reply	Nullable
ingest_ts	timestamp	Ingest time	Add timestamp column

Products

Column Name	Data Type	Description	Action
product_id	string	Product id	PK
product_category_name	string	Category	Join translation
product_name_length	int	Length	Validate
product_description_length	int	Length	Validate
product_photos_qty	int	Count	Validate
product_weight_g	int	Weight grams	Validate
product_length_cm	int	Length cm	Validate
product_height_cm	int	Height cm	Validate
product_width_cm	int	Width cm	Validate
ingest_ts	timestamp	Ingest time	Add timestamp column

Sellers

Column Name	Data Type	Description	Action
seller_id	string	Seller id	PK
seller_zip_code_prefix	int	ZIP prefix	Keep
seller_city	string	City	SCD-2
seller_state	string	State	SCD-2
ingest_ts	timestamp	Ingest time	Add timestamp column

Geolocation

Column Name	Data Type	Description	Action
geolocation_zip_code_prefix	int	ZIP prefix	Reference
geolocation_lat	double	Latitude	Keep
geolocation_lng	double	Longitude	Keep
geolocation_city	string	City	Normalize
geolocation_state	string	State	Normalize
ingest_ts	timestamp	Ingest time	Add timestamp column

category_translation

Column Name	Data Type	Description	Action
product_category_name	string	Portuguese name	Join
product_category_name_english	string	English name	Use

LAYER_WISE_TABLE_LIST :-

Source Dataset	Bronze	Silver	Gold
customers	✓	✓	dim_customer
orders	✓	✓	fact_orders
order_items	✓	✓	fact_orders
payments	✓	✓	fact_orders
reviews	✓	✓	fact_orders
products	✓	✓	dim_product
sellers	✓	✓	dim_seller
geolocation	✓	✓	dim_geolocation (optional)
category_translation	✓	✓	dim_category

FACT_TABLE :-

Column Name	Data Type	Description
order_id	string	Order identifier
product_id	string	Product FK
customer_id	string	Customer FK
seller_id	string	Seller FK
date_id	int	Date FK
order_status	string	Delivered / shipped
price	double	Product price
freight_value	double	Shipping cost
payment_value	double	Revenue
review_score	int	Customer rating
delivery_days	int	SLA metric

DIM_TABLES :-

Table	SCD Type	Reason
dim_customer	SCD-1	No historical tracking needed
dim_product	SCD-1	Corrections overwrite
dim_seller	SCD-2	Location history required
dim_category	SCD-0	Static
dim_date	Derived	Calculated

BUSINESS SCENARIOS :-

1. Leadership wants to monitor daily revenue trends to detect drops or spikes.
2. Finance team tracks MoM OR MONTHLY REVENUE growth to assess business health.
3. Marketplace team wants to reward top sellers.(Top Sellers Performance)
4. Seller Performance Over Time (SCD-2)

   Scenario:

   Seller changed location → delivery performance impacted.

   Why SCD-2:

   Historical seller location matters.
5. Product team wants to know which categories drive revenue(Category Revenue Analysis)
6. Business wants to discontinue or promote weak categories.(Low Performing Categories)
7. Identify sellers causing late deliveries.
8. Do late deliveries reduce ratings? (Delivery Impact on Ratings
9. Regional managers track city/state performance.(City-Wise Revenue)
10. Retention team tracks repeat customers.(Customer Repeat Rate)
11. Finance team wants to understand installments & payment patterns.
12. Marketing wants to target premium customers.(Revenue per customer)
13. Identify sellers likely to churn or get penalized.(Seller Risk Identification)
14. Business wants to plan festival promotions.(Seasonal Sales Analysis)
15. Which products should be promoted or discontinued?(Product Performance)

STEPS

BRONZE LAYER – RAW INGESTION

Store **raw source data** exactly as received.

Steps to Perform (Bronze)

1. Create Bronze folder / schema
2. Read source files (CSV / JSON)
3. Do **NOT** change any data
4. Add ingestion timestamp column
5. Write data in **DELTA format or PARQUET OR CSV**
6. Use **append mode**
7. Do NOT deduplicate
8. Do NOT filter rows
9. Do NOT join tables
10. Keep column names as-is

SILVER LAYER – CLEAN & STANDARDIZE

Create **clean, reliable, standardized datasets**

Steps to Perform (Silver)

1. Read data from **Bronze (Delta)**
2. Rename columns (snake_case)
3. Standardize column names
4. Cast data types correctly

5. Handle **null values**
6. Remove duplicate records
7. Trim spaces from string columns
8. Convert text to lowercase where needed
9. Validate numeric ranges (price, qty)
10. Filter invalid records
11. Apply basic business rules
12. Write cleaned data in **DELTA format**
13. Use **overwrite mode**
14. Partition if required (date-based)
15. Add audit columns (updated_ts)
16. Perform data quality checks
17. Store error records (optional)

GOLD LAYER – BUSINESS READY

Build **analytics-ready fact & dimension tables**

Steps to Perform (Gold)

1. Read required **Silver tables**
2. Identify business grain
3. Create **dimension tables**
4. Apply **SCD logic** where required
5. Create **fact tables**
6. Perform necessary joins

7. Calculate business metrics
8. Validate data consistency
9. Optimize tables for analytics
10. Write in **DELTA format**
11. Use **append mode for fact**
12. Partition fact tables by date
13. Create star schema
14. Validate KPI outputs

SCD IMPLEMENTATION STEPS

SCD-1 (Overwrite)

1. Identify table (Customer, Product)
2. Use business key
3. Compare source vs target
4. Overwrite old values
5. No history preserved

SCD-2 (History Tracking)

1. Identify table (Seller)
2. Add effective_from
3. Add effective_to
4. Add is_current flag
5. Detect attribute changes
6. Expire old record
7. Insert new record

8. Maintain full history

FACT & DIMENSION CREATION STEPS

Dimension Tables

1. Select descriptive columns
2. Remove duplicates
3. Apply SCD logic if needed
4. Validate uniqueness

Fact Tables

1. Identify fact grain
2. Join required silver tables
3. Include foreign keys
4. Add measures (revenue, qty)
5. Validate row count
6. Handle optional relationships
7. Append new records only
8. Optimize for queries

UNIT TESTING / DATA QUALITY STEPS

1. Row count validation
2. Null check on PK columns
3. Duplicate record check
4. Schema validation

FUNCTION & UTILITY NOTEBOOKS

1. Read data function
2. Write Delta function
3. Deduplication function
4. Null handling function
5. Schema validation function
6. If u want create other functions also as per requirements.

DATASETS DOWNLOAD LINK –

https://drive.google.com/drive/u/0/folders/1TutBZXA4WZYUhXeTRrZEvQXDukme81IW